P0 TO P1 PROGRESS

Ordered 1D Empirical-Transition MH for Direct-Permutation TSP Optimization

This report uses an end-to-end comparison. The binary p-bit baseline follows the one-hot QUBO/Ising route. The P-dit method uses the new direct-permutation encoding and the ordered 1D empirical-transition MH kernel. The objective is to compare the complete implementable methods, not to keep the original encoding fixed.

100%	24/24	13/24	1.37e-16
ordered-route validity	8-city exact hits	12-city reference hits	max relative DB error

Scope: P0 direct categorical P-dit, measured sMTJ transition-driven MH, P1 state-geometry analysis, and the new kernel on 8-city and 12-city TSP instances relative to binary p-bits.

Dataset: selected_curve | K=6 | Report date: 2026-07-24

1. Executive Summary

The decisive TSP improvement is not obtained by replacing the random source inside the same one-hot Ising model. It comes from changing both the **state representation** and the **update kernel**: the route is stored directly as a permutation, while a measured device transition k-to-l proposes a local transposition through $q_{kl}=r_k P_{kl}$, followed by a Hastings correction.

Development path

Stage	Problem addressed	Main output
P0-A	Remove invalid Gray/one-hot states and penalties	Direct categorical P-dit; every represented state is valid
P0-B	Use measured device dynamics as proposals	$q_{kl}=r_k P_{kl}$ plus MH correction
P1	Identify the geometry supported by the K=6 device	Ordered 1D locality=0.708 versus Gray locality=0.363
TSP	Eliminate the one-hot feasibility bottleneck	Direct permutation plus reversible cyclic transpositions

Interpretation of the binary p-bit comparison

- **Main comparison:** binary p-bit/one-hot Ising versus ordered 1D direct-permutation MH.

- **Not the main result:** the fixed-one-hot P1 ablation that only changes the Gray/1D embedding. That experiment remains a diagnostic showing that a random-primitive swap alone does not remove the TSP representation bottleneck.

- **Attribution boundary:** the observed gain combines encoding compression, feasible-space reconstruction, proposal locality, and the MH kernel. It cannot be assigned to geometry alone.

The current ordered 1D empirical-transition MH path closes the loop between device physics, state geometry, and TSP constraints: the K=6 state is an actual proposal state, and every route update stays inside the legal permutation space.

2. P0: From Encoding Overhead to Direct Categorical P-dits

P0 first established that direct multivalued variables can replace Gray or one-hot expansions. In the 14-item, 3-partition validation problem, each item stores one category z_i in {0,1,2} and is updated through a softmax conditional distribution. No invalid-code handling or one-hot penalty is required.

14	28	42	90%
direct P-dit variables	Gray binary variables	one-hot binary variables	ground hit at 2048 clocks

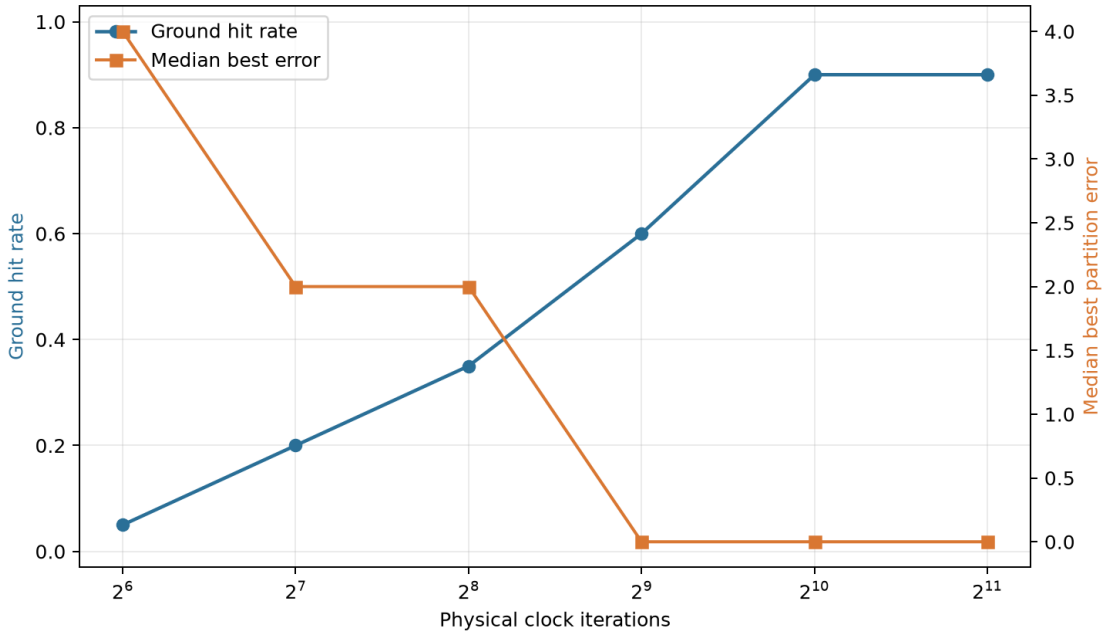


Figure 1. P0 direct P-dit partition solver. The ground-hit rate reaches 90% by 1024 physical-clock iterations; the median best error is already zero at 512 iterations.

Physical-data processing correction completed in P0

- Duration-proportional integer apportionment preserves all 135 true boundaries across 136 dwell segments.

- The new resampling method produces 0 phantom transitions, avoiding artifacts introduced by uniform index resampling.

- This stage proved that direct multivalued states remove invalid encoding regions, but it did not yet determine the correct geometry for the measured K=6 states.

3. P0: From Direct P-dits to Measured-Transition MH

A four-solver experiment then separated representation effects from physical-proposal effects. Solver D introduced a direct categorical proposal based on the measured $q_{kl}=r_k P_{kl}$, with a Metropolis-Hastings correction for asymmetric forward and reverse rates.

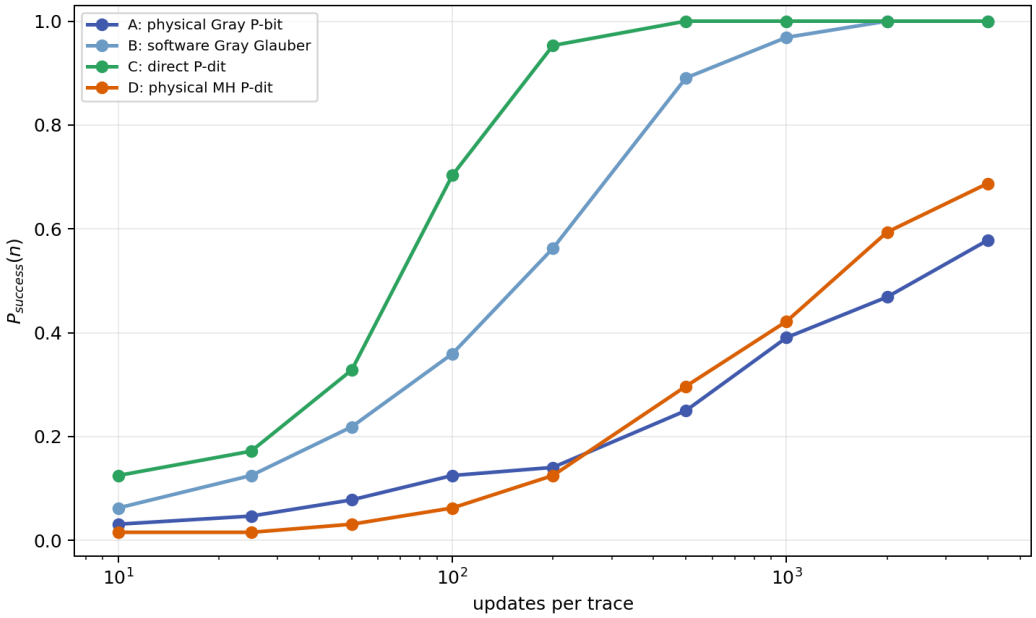


Figure 2. P0 four-solver success curves. C is the software direct P-dit; D is the physical empirical-transition MH P-dit; A and B are Gray-binary paths. This validates the design path but is not the final TSP performance comparison.

Solver	Success at 4000 updates	Median first hit	Valid-state rate
A: physical Gray P-bit	57.8%	579.0	100%
B: software Gray Glauber	100.0%	171.0	100%
C: direct P-dit	100.0%	71.0	100%
D: physical MH P-dit	68.8%	603.5	100%

Design constraints established at this stage

- The MH acceptance ratio must include q_{lk}/q_{kl}; otherwise, asymmetric measured device transitions alter the target stationary distribution.

- A direct categorical representation guarantees validity, but the proposal geometry must still match the device's dominant transition directions.

- P1 therefore focused on identifying which geometry is actually supported by the K=6 transition graph.

4. P1: The Measured K=6 Transition Graph Favors Ordered 1D

The six selected_curve states are naturally ordered by their resistance/conductance means. P1 compared the measured proposal-rate matrix with Gray-coded binary, ordered 1D, and isotropic six-state geometries. Strong transitions concentrate between nearby ordered states rather than forming an isotropic all-to-all graph.

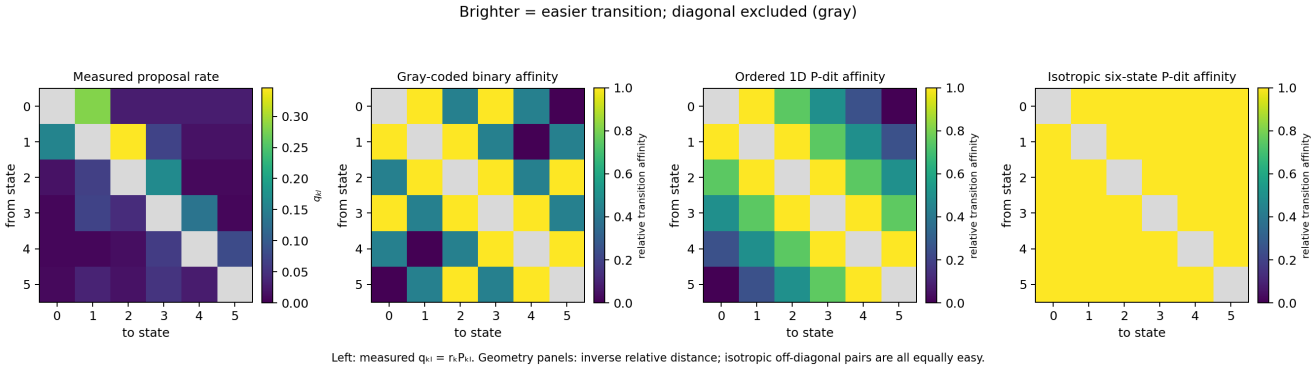


Figure 3. Left: measured q matrix. Remaining panels: relative affinity under each geometry. The banded ordered-1D structure better matches the measured strong transitions; the isotropic model treats every off-diagonal pair as equally distant.

Geometry	Proposal locality	q-weighted normalized distance	Rate vs -distance Pearson
Gray binary	0.363	0.637	0.553
Ordered 1D	**0.708**	**0.292**	**0.645**
Isotropic six-state	0.000	1.000	Undefined: all distances are equal

Why could the isotropic P-dit avoid one-hot encoding while the earlier 1D path could not? The limitation was not one-dimensionality itself. The earlier implementation retained N-squared one-hot TSP variables. The new direct-permutation mapping removes one-hot encoding while preserving the device's measured local ordering.

5. P1: Empirical-Transition MH Sampling Validation

The enumerated validation problem uses four variables, six states, 64 traces per geometry, and 8000 updates per trace. All geometries use the same q_{kl} proposal and MH framework; only the energy geometry changes. Ordered 1D achieves both higher event acceptance and shorter energy autocorrelation.

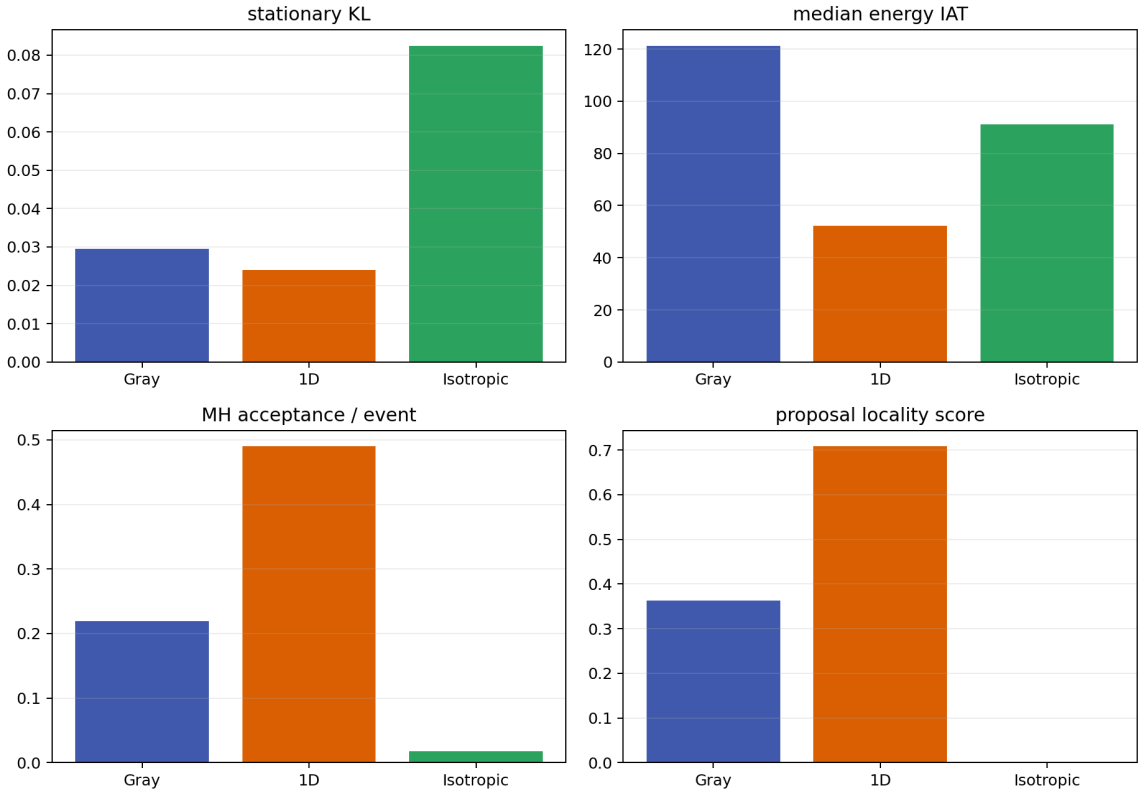


Figure 4. P1 sampling metrics. Ordered 1D reaches an MH acceptance per event of about 0.490 and a median energy IAT of about 52.3 updates. Gray reaches about 0.219 and 121.3, respectively.

Geometry	Stationary KL	MH acceptance/event	Median IAT	Success
Gray-coded binary	0.0295	0.219	121.3	100.0%
Ordered 1D P-dit	0.0239	0.490	52.3	100.0%
Isotropic six-state P-dit	0.0825	0.017	91.1	96.9%

Ordered 1D is not guaranteed to be fastest for every objective. It does, however, use the current K=6 device's local transition structure most faithfully. This supports mapping device transition distance to route-transposition distance in TSP.

6. TSP Kernel: Direct Permutations plus Ordered 1D Empirical-Transition MH

State and proposal

Route state: pi=[0, pi_1, ..., pi_{N-1}], with city 0 fixed to remove rotational degeneracy. A device event k-to-l occurs at $q_{kl}=r_k P_{kl}$. |l-k| is mapped to a cyclic transposition distance outside the fixed origin.

Acceptance rule

$$A[(pi,k) \text{ to } (pi',l)] = \min\{1, \exp[-\beta(L(pi')-L(pi))] * q_{lk}/q_{kl}\}$$

How the kernel eliminates the one-hot bottleneck

Property	Binary p-bit / one-hot Ising	New ordered 1D permutation MH
State size	N-squared binary variables	N route entries; only (N-1)! fixed-origin routes
Validity	Approximated by row/column penalties	Every transposition remains a permutation
Device role	Random update or bias source	k-to-l is the actual q_{kl} proposal state
Reversibility	Defined by the Ising update rule	Same swap for an unordered state pair; MH corrects q asymmetry

Kernel validation

630 detailed-balance flux cases	**1.37e-16** max relative flux error	**PASS** route move reversible under k/l swap	**PASS** adjacent device edges bidirectional

- Distance-1 swaps have positive probability, and adjacent transpositions generate every fixed-origin permutation.

- An accepted move updates both route and device state; a rejection leaves both unchanged, preserving detailed balance in the joint chain.

7. End-to-End TSP Performance: New Encoding and Kernel vs Binary P-bits

This is the requested complete-method comparison, not a one-factor ablation. The binary baseline uses one-hot QUBO/Ising. The P-dit method simultaneously uses direct-permutation encoding and ordered 1D empirical-transition MH. Each method has 24 runs per city size. The saved experiment happens to use 121,500 algorithmic update attempts, but the meaning of an update differs between the two paths.

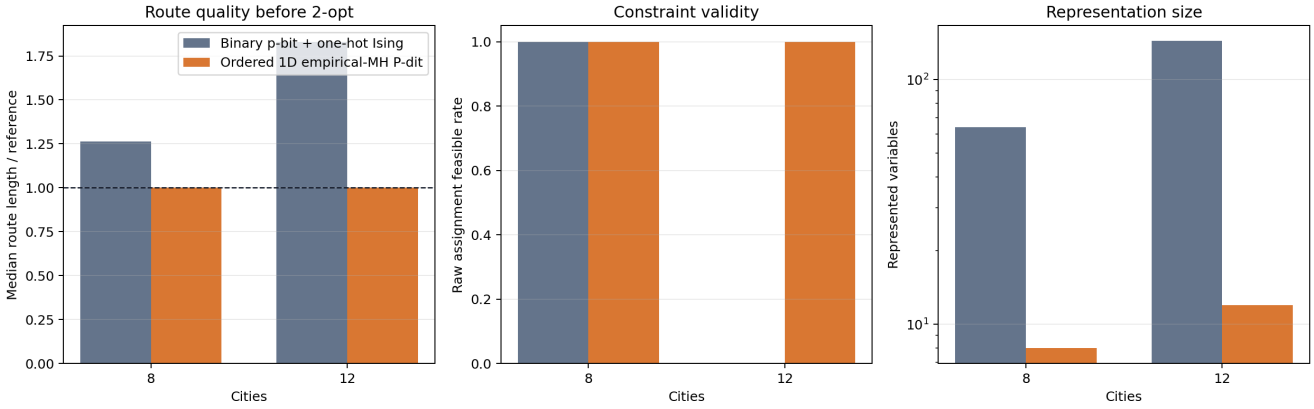


Figure 5. Main TSP comparison. Left: route quality before 2-opt; center: raw-assignment validity; right: representation size. The binary baseline is labeled 'Binary p-bit + one-hot Ising'.

Cities	Method	Variables	Raw feasible	Median route/ref	Raw reference hit
8	Binary p-bit / one-hot	64	100.0%	1.263	0.0%
8	Ordered 1D permutation MH	8	100.0%	1.000	100.0%
12	Binary p-bit / one-hot	144	0.0%	1.831	0.0%
12	Ordered 1D permutation MH	12	100.0%	1.000	54.2%

Key readings

- **8 cities:** ordered MH uses 8 variables and reaches the exact reference in 24/24 raw runs. The binary one-hot path uses 64 variables, has zero raw reference hits, and a median route/ref of 1.263.

- **12 cities:** ordered MH uses 12 variables, maintains 100% raw feasibility, and reaches the reference in 13/24 runs. The binary one-hot path uses 144 variables, has 0% raw feasibility, and a median one-hot violation of 10.5.

- The 12-city reference is obtained by nearest-neighbor all-start plus 2-opt and is not guaranteed to be globally optimal. A ratio of 1 means that this reference was reached.

8. Route Example, Sources of Improvement, and Boundaries

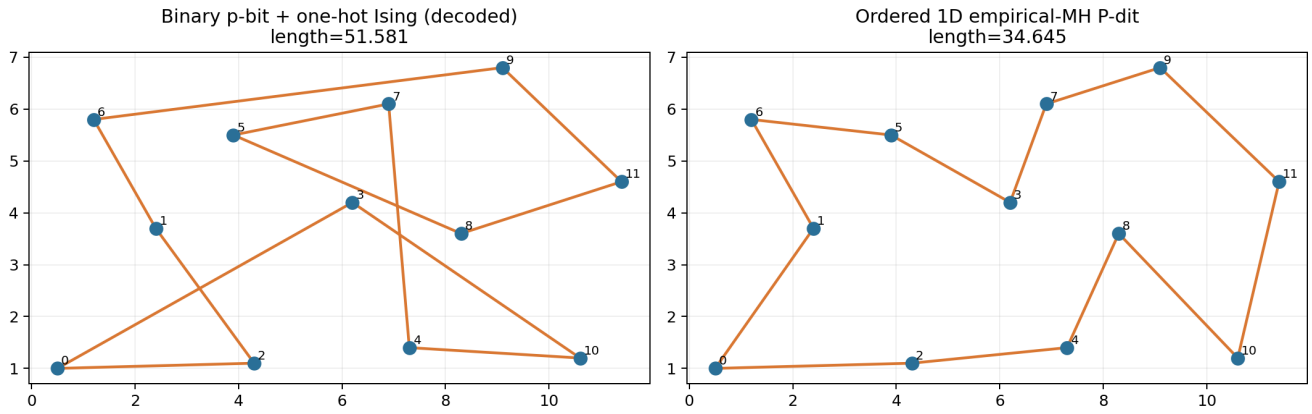


Figure 6. Best saved 12-city routes. Left: decoded binary one-hot route, length 51.581. Right: ordered 1D empirical-MH route, length 34.645, reaching the current reference.

Three layers contribute to the gain

Layer	Mechanism	Observed effect
Representation	N-squared one-hot to N-entry permutation	8x/12x fewer variables for 8/12 cities; search remains legal
Geometry	|l-k| controls transposition distance	Accepted mean distance is about 1.10, preserving local search
Dynamics	Measured q_{kl} proposal plus Hastings correction	Keeps device asymmetry while preserving target detailed balance

Limitations and correct interpretation

- This is an end-to-end algorithm comparison, not a hardware-latency or energy-efficiency comparison. Python wall time does not predict future sMTJ-array speed.

- Update semantics differ: the one-hot path updates one Ising spin; the ordered path advances one empirical proposal-clock tick. Software runtime alone should not be used to rank device performance.

- There are 24 runs per cell. The 8-city reference is exact; the 12-city reference is heuristic. Larger-scale claims require more instances, random seeds, and confidence intervals.

- The P1 fixed-encoding experiment remains useful for geometry diagnosis, but it is not used for the TSP performance claim in this report.

9. Current Conclusion and Next Steps

Current conclusion: ordered 1D empirical-transition MH now forms a complete path from measured K=6 transition statistics to TSP route proposals. Direct-permutation encoding removes the one-hot feasibility bottleneck structurally, while measured-q proposals and the Hastings correction provide both device consistency and theoretical reversibility.

Recommended validation priorities

Priority	Work	Purpose
1	Extend to 16/20/30 cities, multiple instances, and multiple seeds	Test whether the representation advantage remains stable with scale
2	Report success vs clock, proposal events, and accepted swaps	Prevent ambiguity from different update semantics
3	Add adaptive beta and a distance-aware proposal mixture	Balance local refinement and long-range escape
4	Build parallel device mapping plus latency/energy models	Translate algorithmic gains into hardware metrics

Reproduction and figure sources

P0: src/mtj_p0_direct_pdit_partition.py, src/mtj_p0_four_solver_benchmark.py
P1: src/mtj_p1_empirical_transition_mh.py
TSP: src/benchmarks/tsp_binary_pbit_vs_ordered_1d_mh.py
Figures 1-6 are embedded directly from the PNG files saved by these scripts under results/. No plot data were redrawn or modified.

Reference

P-dits: Probabilistic d-dimensional bits for extended-variable probabilistic computing. DOI: 10.1103/4ngx-cmz7

One-sentence summary: the TSP advance comes from the combination of ordered 1D device geometry, empirical-transition MH, and permutation-preserving encoding - not from replacing the random source inside the original one-hot model.